

Lemma 3.7. *Given the assumptions of Theorem 1.4, we have*

$$\mathbb{E}[|\hat{\sigma}_{\mathcal{X}}^2(s, t) - st\sigma_{\mathcal{X}}^2|^2] \rightarrow 0, \quad m \rightarrow \infty \text{ and } \mathbb{E}[|\hat{\sigma}_{\mathcal{Y}}^2(s, t) - st\sigma_{\mathcal{Y}}^2|^2], \quad n \rightarrow \infty$$

for all $s, t \in [0, 1]$.

Proof. We make use of Corollary 3.8.

$$\begin{aligned} & \mathbb{E}[|\hat{\sigma}_{\mathcal{X}}^2(s, t) - st\sigma_{\mathcal{X}}^2|^2] \\ &= \frac{1}{4} \mathbb{E}\left[\left(\frac{1}{m(m-1)} \sum_{i=1}^{\lfloor ms \rfloor} \sum_{\substack{j=1 \\ j \neq i}}^{\lfloor mt \rfloor} W_r^2(\text{PD}(\mathcal{X}_i), \text{PD}(\mathcal{X}_j)) - st\mathbb{E}[W_r^2(\text{PD}(\mathcal{X}_0), \text{PD}(\mathcal{X}'))]\right)^2\right] \\ &\lesssim n^{-1} \end{aligned}$$

The result holds in the same spirit for $\hat{\sigma}_{\mathcal{Y}}^2(s, t)$. $\square$

Proof of Theorem 1.4. Again the proof splits in four major steps and is very similar to the proof of Theorem 3.1.

Step 1. In the first step we verify that the conditions of Assumption 2.1 (p, q, ρ) are satisfied in case (a) and case (b).

Case (a). If $\#\mathcal{X}_i, \#\mathcal{Y}_j$ are uniformly bounded with probability 1 for all $i, j \in \mathbb{Z}$ the kernel in (1.13) satisfies part (A1) in Assumption 2.1 (p, q, ρ) for $\rho = 1 - 1/r$ and any $q \geq 1$ by Lemma 3.6 part (B).

Case (b). The kernel satisfies part (A1) in Assumption 2.1 (p, q, ρ) for any $q \geq 1$ and each $\rho < 1 - d/r$ by Lemma 3.6 part (A).

Thus, if we choose $p = 1$ and $q = \infty$ in both cases (a) and (b), the data $(\mathcal{X}_i)_i$ and $(\mathcal{Y}_i)_i$ satisfy the approximation condition (A2) in Assumption 2.1 (p, q, ρ) because we require

$$\sum_{m \geq 1} m \mathbb{E}[d_H(\mathcal{X}_0, \mathcal{X}_0^{(m)})^\rho] < \infty \text{ and } \sum_{m \geq 1} m \mathbb{E}[d_H(\mathcal{Y}_0, \mathcal{Y}_0^{(m)})^\rho] < \infty$$

for $\rho = 1 - 1/r$ in case (a) and in case (b) for some $\rho \in (0, 1 - d/r)$ where additionally $d < r$. Hence, we can apply Theorem 2.3 to the processes (1.12) and obtain due to the independence between the samples $(\mathcal{X}_i)_i$ and $(\mathcal{Y}_i)_i$

$$\begin{pmatrix} \sqrt{m}(\hat{\sigma}_{\mathcal{X}}^2(s, t) - st\sigma_{\mathcal{X}}^2) : 0 \leq s, t \leq 1 \\ \sqrt{n}(\hat{\sigma}_{\mathcal{Y}}^2(s, t) - st\sigma_{\mathcal{Y}}^2) : 0 \leq s, t \leq 1 \end{pmatrix} \Rightarrow \begin{pmatrix} t\mathcal{W}_{\Gamma_{\mathcal{X}}}(s) + s\mathcal{W}_{\Gamma_{\mathcal{X}}}(t) : 0 \leq s, t \leq 1 \\ t\mathcal{W}_{\Gamma_{\mathcal{Y}}}(t) + s\mathcal{W}_{\Gamma_{\mathcal{Y}}}(s) : 0 \leq s, t \leq 1 \end{pmatrix} \quad (3.24)$$

in the product space $D_2 \times D_2$, where $D_2 = D([0, 1]^2)$. Note that the Wiener processes on the right-hand side $\mathcal{W}_{\Gamma_{\mathcal{X}}}, \mathcal{W}_{\Gamma_{\mathcal{Y}}}$ are necessarily independent. Thus, we can characterize the Wiener processes as $\mathcal{L}(\mathcal{W}_{\Gamma_{\mathcal{X}}}) = \mathcal{L}(\sqrt{\Gamma_{\mathcal{X}}}B_1)$ and $\mathcal{L}(\mathcal{W}_{\Gamma_{\mathcal{Y}}}) = \mathcal{L}(\sqrt{\Gamma_{\mathcal{Y}}}B_2)$, where B_1 and B_2 denote two independent Brownian motions on $[0, 1]$.

Step 2. Now, we can consider the joint behavior of two dimensional random vector given on the left-hand side of (1.15). The first component is

$$\sqrt{m+n}(\hat{D}_{m,n}^2 - D^2) = \sqrt{m+n}(\hat{D}_{m,n} - D)(\hat{D}_{m,n} + D)$$

$$\begin{aligned}
&= \left(\sqrt{\frac{m+n}{m}} \sqrt{m} (\hat{\sigma}_{\mathcal{X}}^2(1,1) - \sigma_{\mathcal{X}}^2) - \sqrt{\frac{m+n}{n}} \sqrt{n} (\hat{\sigma}_{\mathcal{Y}}^2(1,1) - \sigma_{\mathcal{Y}}^2) \right) \\
&\quad \times (\hat{\sigma}_{\mathcal{X}}^2(1,1) - \hat{\sigma}_{\mathcal{Y}}^2(1,1) + \sigma_{\mathcal{X}}^2 - \sigma_{\mathcal{Y}}^2) =: Z_{m,n}.
\end{aligned} \tag{3.25}$$

The second component is an integral and the square root of its integrand is

$$\begin{aligned}
&\sqrt{m+n} (\hat{D}_{m,n}^2(s,t) \mp (stD)^2 - (st\hat{D}_{m,n})^2) \\
&= \sqrt{m+n} (\hat{D}_{m,n}(s,t) - stD) (\hat{D}_{m,n}(s,t) + stD) \\
&\quad - (st)^2 \sqrt{m+n} (\hat{D}_{m,n} - D) (\hat{D}_{m,n} + D) \\
&= \left(\sqrt{\frac{m+n}{m}} \sqrt{m} (\hat{\sigma}_{\mathcal{X}}^2(s,t) - st\sigma_{\mathcal{X}}^2) - \sqrt{\frac{m+n}{n}} \sqrt{n} (\hat{\sigma}_{\mathcal{Y}}^2(s,t) - st\sigma_{\mathcal{Y}}^2) \right) \\
&\quad \times (\hat{\sigma}_{\mathcal{X}}^2(s,t) + st\sigma_{\mathcal{X}}^2 - \hat{\sigma}_{\mathcal{Y}}^2(s,t) - st\sigma_{\mathcal{Y}}^2) \\
&\quad - (st)^2 \sqrt{m+n} (\hat{D}_{m,n} - D) (\hat{D}_{m,n} + D) =: \tilde{Z}_{m,n}(s,t).
\end{aligned} \tag{3.26}$$

Note that the last term on the right-hand side of (3.26) is the $(st)^2$ -multiple of the left-hand side of (3.25).

Step 3. Next, we prove the desired convergence, which essentially is an application of the continuous mapping theorem. To this end, we define appropriate mappings F_1, F_2 on $D_2 \times D_2$.

We begin with the first coordinate which is given in (3.25) and define the functional

$$F_1 : D_2 \times D_2 \rightarrow \mathbb{R}, \quad (u, v) \mapsto \left(\frac{u(1,1)}{\sqrt{\tau}} - \frac{v(1,1)}{\sqrt{1-\tau}} \right) (2\sigma_{\mathcal{X}}^2 - 2\sigma_{\mathcal{Y}}^2).$$

Given two continuous functions u, v in D_2 , F_1 is continuous at (u, v) because the point evaluation is a continuous functional at continuous functions in the Skorokhod space.

Moreover, $m/(m+n) \rightarrow \tau$ as well as $\hat{\sigma}_{\mathcal{X}}^2 \rightarrow \sigma_{\mathcal{X}}^2$ and $\hat{\sigma}_{\mathcal{Y}}^2 \rightarrow \sigma_{\mathcal{Y}}^2$ in probability by Lemma 3.7. And all in all, it follows with some calculations that

$$Z_{m,n} - F_1(\sqrt{m}(\hat{\sigma}_{\mathcal{X}}^2(1,1) - \sigma_{\mathcal{X}}^2), \sqrt{n}(\hat{\sigma}_{\mathcal{Y}}^2(1,1) - \sigma_{\mathcal{Y}}^2)) \rightarrow 0 \tag{3.27}$$

in probability.

Moreover, as the limits on the right-hand side of (3.24) are continuous elements of D_2 , we can apply the continuous mapping theorem to F_1

$$\begin{aligned}
&F_1(\sqrt{m}(\hat{\sigma}_{\mathcal{X}}^2(1,1) - \sigma_{\mathcal{X}}^2), \sqrt{n}(\hat{\sigma}_{\mathcal{Y}}^2(1,1) - \sigma_{\mathcal{Y}}^2)) \\
&\Rightarrow F_1((tW_{\Gamma_{\mathcal{X}}}(s) + sW_{\Gamma_{\mathcal{X}}}(t))_{s,t}, (tW_{\Gamma_{\mathcal{Y}}}(s) + sW_{\Gamma_{\mathcal{Y}}}(t))_{s,t}) \\
&= \left(\frac{2}{\sqrt{\tau}} W_{\Gamma_{\mathcal{X}}}(1) - \frac{2}{\sqrt{1-\tau}} W_{\Gamma_{\mathcal{Y}}}(1) \right) (2\sigma_{\mathcal{X}}^2 - 2\sigma_{\mathcal{Y}}^2).
\end{aligned}$$

Next, we consider $\tilde{Z}_{m,n}$ which is given in (3.26). Define the operator

$$F_2 : D_2 \times D_2 \rightarrow D_2,$$

$$(u, v) \mapsto \left(2st \left\{ \frac{u(s, t)}{\sqrt{\tau}} - \frac{v(s, t)}{\sqrt{1-\tau}} \right\} (\sigma_{\mathcal{X}}^2 - \sigma_{\mathcal{Y}}^2) - (st)^2 F_1(u, v) \right)_{s,t}.$$

Now, it follows with some calculations that

$$d_2 \left(\tilde{Z}_{m,n}, F_2([\sqrt{m}(\hat{\sigma}_{\mathcal{X}}^2(s, t) - st\sigma_{\mathcal{X}}^2)]_{s,t}, [\sqrt{n}(\hat{\sigma}_{\mathcal{Y}}^2(s, t) - st\sigma_{\mathcal{Y}}^2)]_{s,t}) \right) \rightarrow 0 \text{ in probability.} \quad (3.28)$$

Observe that F_2 is discontinuous only on $\text{Dis}(F_2) = D_2 \setminus \mathcal{C}_2 \times D_2 \setminus \mathcal{C}_2$, where $\mathcal{C}_2$ denotes the set of continuous functions on $[0, 1]^2$. Now,

$$\mathbb{P}([tW_{\Gamma_X}(s) + sW_{\Gamma_X}(t)]_{s,t}, [tW_{\Gamma_Y}(s) + sW_{\Gamma_Y}(t)]_{s,t} \in \text{Dis}(F_2)) = 0.$$

Consequently, an application of the continuous mapping theorem yields

$$\begin{aligned} & F_2([\sqrt{m}(\hat{\sigma}_{\mathcal{X}}^2(s, t) - st\sigma_{\mathcal{X}}^2)]_{s,t}, [\sqrt{n}(\hat{\sigma}_{\mathcal{Y}}^2(s, t) - st\sigma_{\mathcal{Y}}^2)]_{s,t}) \\ & \Rightarrow F_2([tW_{\Gamma_X}(s) + sW_{\Gamma_X}(t)]_{s,t}, [tW_{\Gamma_Y}(s) + sW_{\Gamma_Y}(t)]_{s,t}) \\ & = \left(2st \left\{ \frac{tW_{\Gamma_X}(s) + sW_{\Gamma_X}(t)}{\sqrt{\tau}} - \frac{tW_{\Gamma_Y}(s) + sW_{\Gamma_Y}(t)}{\sqrt{1-\tau}} \right\} (\sigma_{\mathcal{X}}^2 - \sigma_{\mathcal{Y}}^2) \right. \\ & \quad \left. - 4(st)^2 \left(\frac{W_{\Gamma_X}(1)}{\sqrt{\tau}} - \frac{W_{\Gamma_Y}(1)}{\sqrt{1-\tau}} \right) (\sigma_{\mathcal{X}}^2 - \sigma_{\mathcal{Y}}^2) \right)_{s,t} \end{aligned}$$

in D_2 . The law of the last right-hand side equals

$$\begin{aligned} & \left(\left\{ \sqrt{\frac{\Gamma_X}{\tau}}(tB_1(s) + sB_1(t)) - \sqrt{\frac{\Gamma_Y}{1-\tau}}(tB_2(s) + sB_2(t)) \right\} \times 2st(\sigma_{\mathcal{X}}^2 - \sigma_{\mathcal{Y}}^2) - 2(st)^2 \xi B(1) \right)_{s,t} \\ & \stackrel{\mathcal{L}}{=} \left(\xi(st(tB_1(s) + sB_1(t) - 2stB_1(1))) \right)_{s,t}. \end{aligned}$$

Moreover, set

$$G: D_2 \rightarrow \mathbb{R}, \quad u \mapsto \left\{ \int_{[0,1]^2} u(s, t)^2 \nu(d(s, t)) \right\}^{1/2}.$$

Since taking the pointwise square, integration and the square root are $\mathbb{P}_{(B_1, B_2)}$ -a.s. continuous functionals on D_2 , the composition $G \circ F_2$ is an $\mathbb{P}_{(B_1, B_2)}$ -a.s. continuous function. Thus, using the result from (3.28), we have on the one hand

$$\begin{aligned} & \sqrt{m+n} \hat{V}_{m,n} - G \circ F_2([\sqrt{m}(\hat{\sigma}_{\mathcal{X}}^2(s, t) - st\sigma_{\mathcal{X}}^2)]_{s,t}, [\sqrt{n}(\hat{\sigma}_{\mathcal{Y}}^2(s, t) - st\sigma_{\mathcal{Y}}^2)]_{s,t}) \\ & = G(\tilde{Z}_{m,n}) - G \circ F_2([\sqrt{m}(\hat{\sigma}_{\mathcal{X}}^2(s, t) - st\sigma_{\mathcal{X}}^2)]_{s,t}, [\sqrt{n}(\hat{\sigma}_{\mathcal{Y}}^2(s, t) - st\sigma_{\mathcal{Y}}^2)]_{s,t}) \rightarrow 0 \end{aligned} \quad (3.29)$$

in probability.